# *SWIGGY FOOD DELIVERY DATA ANALYSIS & MACHINE LEARNING (2025)*

**End-to-End Data Science & Business Intelligence Case Study**

***Data Engineering • EDA • Statistical Analysis • Feature Engineering • Machine Learning • Customer Segmentation • Business Insights***

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**Project Overview**

This project presents an end-to-end data science and business analytics case study on Swiggy food delivery operations. It focuses on transforming raw transactional data into actionable business intelligence using a structured analytical pipeline involving data preprocessing, exploratory data analysis (EDA), statistical testing, feature engineering, customer segmentation, and predictive modeling.

The study delivers **data-driven insights and strategic recommendations** to improve operational efficiency, customer experience, and revenue optimization in the food delivery ecosystem.

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**Key Objectives**

- Perform data cleaning and preprocessing on real-world food delivery data
- Conduct exploratory data analysis to identify trends and patterns
- Apply statistical hypothesis testing for business validation
- Engineer features to improve analytical and model performance
- Segment customers using K-Means clustering
- Build and evaluate machine learning models for prediction
- Generate actionable business insights and recommendations

**Tools & Technologies**

**Programming:** Python 3.x

**Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly

**Statistical Analysis:** SciPy, Statsmodels

**Machine Learning:** Scikit-learn, Linear Regression, K-Means Clustering, Cross Validation, Feature Scaling

**Environment:** Jupyter Notebook, VS Code, Git, GitHub

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**Project Domain**

Food Delivery Analytics | Business Intelligence | Data Science | Machine Learning

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Swiggy Food Delivery Data Analysis & Machine Learning (2025)
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## Executive Summary

- Swiggy generates large-scale food delivery data that can be used for business insights and decision-making.
- The project performs a complete **end-to-end data science workflow** including:
    - Data cleaning
    - Exploratory Data Analysis (EDA)
    - Statistical analysis
    - Feature engineering
    - Machine learning
    - Customer segmentation
- Data preprocessing ensures the dataset is **clean, consistent, and reliable** for analysis.
- EDA is used to understand:
    - Customer behavior
    - Restaurant performance
    - Pricing trends
    - Cuisine preferences
    - Delivery patterns
- Statistical methods (correlation, hypothesis testing) are used to **validate relationships between variables**.
- **K-Means clustering** is applied to segment customers into meaningful groups based on ordering behavior.
- Machine learning models are built to **predict key business outcomes**, evaluated using MAE, RMSE, and $R^2$ score.
- Feature importance is analyzed to understand **key drivers affecting predictions**.
- The project converts data insights into **business recommendations** for:
    - Delivery optimization
    - Customer satisfaction improvement

  - Restaurant performance enhancement
  - Revenue growth
- Built using Python and libraries like Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, etc.
- The project demonstrates a **complete real-world data science lifecycle** and is suitable for a professional portfolio.
- This documentation is structured to meet **industry-level standards for data science project reporting and GitHub presentation**.
- It serves as a **reproducible and scalable framework** for future analytics and machine learning enhancements.

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# Business Problem

The rapid growth of online food delivery platforms has significantly transformed the food and restaurant industry by offering customers convenient access to a wide variety of cuisines, restaurants, and real-time delivery services. As one of India's leading food delivery platforms, Swiggy handles a massive volume of daily transactions, including customer orders, restaurant listings, delivery operations, and user interactions. This large-scale operational data presents a strong opportunity to extract meaningful insights using data analytics and machine learning techniques.

Despite the availability of rich data, food delivery platforms face several critical business challenges. Customers expect fast and reliable deliveries, affordable pricing, accurate restaurant recommendations, and consistent service quality. Restaurants aim to increase visibility, improve ratings, attract more customers, and maximize revenue. At the same time, delivery partners need optimized routing and efficient time management, while the platform itself focuses on improving operational efficiency, customer satisfaction, and profitability.

However, in the absence of structured data analysis, it becomes difficult to identify the key factors influencing customer behavior, restaurant performance, delivery efficiency, and pricing strategies. Business decisions made without data-driven insights may lead to inefficiencies, poor customer experience, and suboptimal resource allocation.

The core business problem addressed in this project is the transformation of raw Swiggy operational data into actionable business intelligence. This involves systematic data cleaning, exploratory data analysis, statistical evaluation, and machine learning techniques to uncover hidden patterns and relationships within the dataset. The goal is to understand what drives customer satisfaction, what impacts delivery performance, and how restaurant characteristics influence overall business outcomes.

This project further applies advanced analytics techniques such as K-Means clustering for customer segmentation and regression modeling for predictive analysis. These methods help identify distinct customer groups, predict key business metrics, and understand the most influential factors affecting platform performance.

# Business Objectives

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The objective of this project is to apply data science and machine learning techniques to analyze Swiggy's food delivery data and extract actionable insights for improving business performance, customer experience, and operational efficiency.

The project follows an end-to-end data science workflow including data cleaning, exploratory data analysis, statistical testing, feature engineering, machine learning modeling, and business interpretation.

**Primary Business Objectives**

- Understand Swiggy's food delivery ecosystem using real-world data.
- Ensure data quality by handling missing values, duplicates, and outliers.
- Analyze customer behavior, restaurant performance, pricing, ratings, and delivery patterns.
- Identify key operational trends and inefficiencies.
- Engineer meaningful features for analysis and modeling.
- Validate relationships using statistical methods.
- Segment customers or restaurants using clustering techniques.
- Build predictive models for key business metrics.
- Identify major drivers of business performance.
- Convert insights into actionable recommendations.

**Analytical Objectives**

- Identify factors affecting delivery time and customer satisfaction.
- Analyze impact of pricing, ratings, and cuisine on orders.
- Discover patterns in customer behavior.
- Identify meaningful customer/restaurant segments.
- Determine key predictive features.
- Study relationships between operational and financial variables.
- Improve forecasting using machine learning models.

**Operational Objectives**

- Reduce delivery delays and inefficiencies.

- Improve restaurant and delivery partner performance.
- Optimize resource allocation.
- Enhance demand forecasting accuracy.
- Identify high and low-performing segments.

**Strategic Objectives**

- Improve customer retention through data insights.
- Optimize pricing and promotional strategies.
- Strengthen restaurant partnerships.
- Support expansion and growth decisions.
- Build a scalable analytics framework.

**Expected Business Outcomes**

- Clean and structured dataset.
- Actionable insights from analysis.
- Customer/restaurant segmentation.
- Predictive models for decision support.
- Data-driven business recommendations.
- Improved efficiency and customer experience.

Overall, this project demonstrates how data science can transform raw data into meaningful business insights that support strategic growth in the food delivery industry.

# Swiggy Business Overview

Swiggy is one of India's leading on-demand convenience platforms offering food delivery, grocery delivery, and quick-commerce services through a technology-driven digital marketplace. Founded in 2014 and headquartered in Bengaluru, Swiggy connects customers, restaurants, grocery stores, cloud kitchens, and delivery partners through a unified platform. By leveraging artificial intelligence, data analytics, and real-time logistics optimization, Swiggy has significantly transformed the food delivery ecosystem and improved customer convenience across urban and semi-urban regions.

Operationally, Swiggy follows a real-time workflow where customer orders are instantly routed to restaurants, while the system simultaneously assigns optimized delivery partners based on location, traffic conditions, and delivery capacity. Once prepared, the order is picked up and delivered with live tracking updates, ensuring transparency and efficiency throughout the process. This technology-enabled system helps reduce delivery time, optimize logistics, and enhance customer satisfaction.

Data plays a central role in Swiggy's ecosystem. Every transaction, customer interaction, and delivery movement generates valuable data that is used for business intelligence and predictive analytics. Swiggy applies data-driven decision-making in areas such as demand forecasting, customer behavior analysis, restaurant performance evaluation, delivery optimization, dynamic pricing, and personalized recommendations.

Despite its success, Swiggy faces several challenges including high delivery costs during peak hours, traffic-related delays, maintaining consistent food quality, managing delivery partner availability, and balancing discounts with profitability.

Data Science is therefore a core enabler for Swiggy's business strategy. It supports applications such as delivery time prediction, customer segmentation, recommendation systems, fraud detection, demand forecasting, and route optimization. By transforming raw data into actionable insights, Swiggy enhances decision-making, improves customer satisfaction, and strengthens its competitive position in the market.

This project demonstrates an end-to-end data science workflow using Swiggy food delivery data, including data cleaning, exploratory data analysis, statistical testing, feature engineering, clustering, and machine learning modeling. The objective is to extract meaningful business insights, understand customer behavior, and provide data-driven recommendations that can improve operational efficiency and business performance. The project also serves as a professional portfolio piece showcasing practical skills in data analysis, machine learning, and business communication.

## Food Delivery Industry Analysis

The online food delivery industry is one of the fastest-growing segments of digital commerce, driven by smartphone adoption, affordable internet, digital payments, and changing urban lifestyles. Consumers increasingly prefer ordering food through mobile applications due to convenience, variety, real-time tracking, and fast delivery services. In India, platforms like Swiggy and Zomato dominate the market, evolving from simple ordering apps into full-scale ecosystems integrating restaurants, delivery partners, payments, and AI-driven recommendation systems.

The industry has evolved significantly over time. Initially, restaurants handled deliveries manually via phone orders, resulting in limited choices and slow service. This shifted to digital aggregator platforms that introduced online menus, reviews, secure payments, and tracking systems. Today, the industry operates as an intelligent ecosystem powered by AI, machine learning, GIS systems, predictive analytics, and route optimization to improve efficiency and customer satisfaction.

Data science plays a critical role in this industry by enabling demand forecasting, customer behavior prediction, restaurant recommendations, delivery route optimization, fraud detection, and personalized marketing. Machine learning models continuously analyze historical order data to improve decision-making and enhance user experience.

Data analytics helps organizations measure performance, track customer satisfaction, analyze sales trends, evaluate restaurant quality, and optimize pricing and promotions. It also supports customer segmentation and demand forecasting, enabling better business planning and resource allocation.

Machine learning applications in food delivery include delivery time prediction, recommendation systems, dynamic pricing, demand forecasting, customer segmentation, fraud detection, churn prediction, and order value prediction. These systems improve efficiency, reduce costs, and enhance customer experience.

This project applies data science techniques to analyze Swiggy food delivery data, including data cleaning, exploratory data analysis, statistical testing, clustering, and predictive modeling. The goal is to extract meaningful business insights, identify customer behavior patterns, and provide actionable recommendations to improve operational efficiency, customer satisfaction, and overall business performance.

## Dataset Description

The dataset used in this project contains transactional and operational data from Swiggy's food delivery platform, including restaurant details, cuisine types, customer ratings, pricing, delivery performance, and order-related attributes. It is used to analyze customer behavior, evaluate restaurant performance, and build predictive models for business decision-making.

The dataset was sourced from a publicly available Swiggy dataset and is used strictly for educational and analytical purposes. It is structured in CSV format and contains both numerical and categorical variables suitable for statistical analysis, visualization, and machine learning tasks such as regression and clustering.

The dataset supports key business objectives such as understanding restaurant performance, analyzing customer satisfaction, studying delivery efficiency, and identifying meaningful customer and restaurant segments. It also enables predictive modeling to estimate business outcomes and generate actionable insights.

From a data perspective, the dataset includes restaurant information (name, cuisine, location), customer experience metrics (ratings, reviews), delivery-related attributes (delivery time, logistics factors), and pricing information (cost, discounts, order value). These features allow a comprehensive analysis of the food delivery ecosystem.

The dataset contains mixed data types, including numerical variables for statistical modeling, categorical variables for segmentation, and text-based fields for restaurant and cuisine descriptions. This makes it suitable for a wide range of analytical techniques including descriptive statistics, hypothesis testing, clustering, and regression modeling.

However, the dataset has certain limitations. Missing values, potential outliers, and limited feature scope may affect analysis accuracy. Additionally, results are dependent on data quality and may not fully generalize to all regions or time periods.

Overall, the Swiggy dataset provides a strong foundation for end-to-end data science analysis, enabling data-driven insights that can improve operational efficiency, customer satisfaction, and strategic business decisions.

# Project Architecture

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The **Swiggy Food Delivery Data Analysis & Machine Learning (2025)** project follows a modular end-to-end data science pipeline that transforms raw operational data into actionable insights and predictive models. The workflow integrates data ingestion, cleaning, feature engineering, EDA, statistical analysis, machine learning, and business reporting in a structured and reproducible manner.

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- **End-to-End Workflow**

Swiggy Dataset

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Data Import

↓

Data Quality Check

(Missing Values • Duplicates • Data Types • Outliers)

↓

Data Cleaning

↓

Feature Engineering

(Encoding • Scaling • Derived Features)

↓

EDA

↓

Statistical Analysis

(Correlation • T-Test • ANOVA • Chi-Square)

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Clustering Regression Models

(K-Means)              (ML Models)

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Model Evaluation

(MAE • RMSE • $R^2$ • Silhouette Score)

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Business Insights

↓

Final Report & GitHub

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- **Architecture Components**

**1. Data Acquisition**

Raw dataset is imported and structured for analysis.

**Input:** CSV file
**Output:** DataFrame

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**2. Data Quality Check**

Ensures dataset reliability.

- Missing values
- Duplicates
- Data types
- Outliers

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**3. Data Cleaning**

Prepares dataset for analysis.

- Missing value handling
- Duplicate removal
- Type correction

- Standardization

**Output:** Clean dataset

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**4. Feature Engineering**

Creates meaningful variables.

- Cost per item
- Peak hour flag
- Weekend orders
- Customer segments
- Delivery categories

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**5. EDA**

Identifies patterns and trends.

- Distribution analysis
- Correlation study
- Category comparisons
- Trend analysis

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**6. Statistical Analysis**

Validates relationships.

- Correlation
- T-Test
- ANOVA
- Chi-Square

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**7. Machine Learning**

Predictive modeling pipeline.

- Train-test split

- Scaling
- Model training
- Cross-validation

**Models:**

- Linear Regression
- Decision Tree
- Random Forest
- Gradient Boosting

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**8. Customer Segmentation**

Unsupervised learning for grouping users.

- K-Means clustering
- Elbow method
- PCA visualization

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**9. Model Evaluation**

**Regression Metrics:**

- MAE
- RMSE
- $R^2$

**Clustering Metrics:**

- Silhouette Score
- Inertia

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**10. Business Intelligence**

Final insights for decision-making.

- Customer behavior
- Delivery performance

- Restaurant analysis
- Revenue insights
- Optimization strategies

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- **Technology Stack**

| **Component** | **Tools** |
|---|---|
| Language | Python |
| Data Handling | Pandas, NumPy |
| Visualization | Matplotlib, Seaborn |
| Statistics | SciPy |
| ML | Scikit-learn |
| Environment | Jupyter, VS Code |
| Version Control | Git, GitHub |

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**Key Benefits**

- Modular and scalable design
- End-to-end reproducible pipeline
- Industry-standard CRISP-DM approach
- Supports analytics + ML workflow
- Easy GitHub deployment
- Strong business interpretability

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# Technology Stack

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The **Swiggy Food Delivery Data Analysis & Machine Learning (2025)** project uses a modern data science stack covering the full pipeline from data processing to model deployment and business insights.

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### Programming Language

**Python 3.11+**

Python is the core language used for all data science operations due to its simplicity and strong ecosystem.

**Used for:**

- Data cleaning
- Data analysis
- Machine learning
- Visualization
- Model evaluation

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### Development Environment

**Jupyter Notebook**

Used for interactive development and analysis.

**Benefits:**

- Step-by-step execution
- Easy visualization
- Reproducible workflow

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### Data Processing Libraries

**Pandas**

- Data loading
- Cleaning

- Feature engineering
- Aggregations

**NumPy**

- Numerical operations
- Array processing
- Mathematical computations

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**Visualization Libraries**

**Matplotlib**

- Basic charts
- Histograms
- Scatter plots

**Seaborn**

- Heatmaps
- Distribution plots
- Statistical graphs

**Plotly**

- Interactive dashboards
- Dynamic visualizations

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**Statistical Libraries**

**SciPy**

- Hypothesis testing
- Statistical analysis
- Probability functions

**Statsmodels**

- Regression analysis
- Statistical modeling

- Diagnostics

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**Machine Learning**

**Scikit-Learn**

**Models used:**

- Linear Regression
- Decision Tree
- Random Forest
- K-Means Clustering

**Techniques:**

- Scaling
- Encoding
- Train-test split
- Cross-validation

**Metrics:**

- MAE
- RMSE
- $R^2$ Score

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**Model Storage**

**Pickle**

Used to save trained models and preprocessing objects for reuse.

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**Version Control**

**Git & GitHub**

- Code tracking
- Collaboration
- Project hosting

- Portfolio management

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**Documentation Tools**

- Markdown
- PDF export
- GitHub README

Used for professional reporting and presentation.

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**Technology Stack Summary**

| Category | Tool | Purpose |
| --- | --- | --- |
| Language | Python | Data Science |
| IDE | Jupyter | Analysis |
| Data Handling | Pandas | Processing |
| Computation | NumPy | Math |
| Visualization | Matplotlib | Charts |
| Visualization | Seaborn | Stats plots |
| Interactive Viz | Plotly | Dashboards |
| Statistics | SciPy | Testing |
| Modeling | Statsmodels | Regression |
| ML Library | Scikit-Learn | ML models |
| Storage | Pickle | Model saving |
| Version Control | Git | Tracking |
| Hosting | GitHub | Portfolio |
| Docs | Markdown/PDF | Reporting |

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**Workflow**

Dataset

↓

Pandas / NumPy

↓

EDA (Matplotlib / Seaborn)

↓

Statistics (SciPy)

↓

Machine Learning (Scikit-Learn)

↓

Evaluation (Metrics)

↓

Insights & Recommendations

↓

GitHub Documentation

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This stack ensures a complete end-to-end data science workflow suitable for real-world industry projects.

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## Data Cleaning

Data cleaning is the process of transforming raw data into a structured and reliable format by handling missing values, removing duplicates, correcting data types, fixing inconsistencies, and treating outliers. It is a critical step in ensuring that the dataset is accurate, consistent, and suitable for analysis and machine learning.

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**Objectives**

- Improve data quality and reliability
- Remove duplicate records
- Handle missing values appropriately
- Correct and standardize data types
- Detect and treat outliers
- Ensure consistency in data formatting
- Prepare dataset for analysis and modeling

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**Data Quality Check**

The dataset was evaluated for key issues:

| Check | Purpose |
|---|---|
| Missing Values | Identify incomplete data |
| Duplicates | Remove repeated records |
| Data Types | Ensure correct format |
| Outliers | Detect abnormal values |
| Consistency | Standardize data |

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**Missing Values**

Missing values were identified using isnull().sum() and handled based on feature type:

- Numerical → mean / median imputation
- Categorical → mode / “Unknown” category

- Time-series (if applicable) → forward/backward fill

This ensures minimal information loss while maintaining data integrity.

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**Duplicate Removal**

Duplicate records were identified and removed to ensure uniqueness of observations. After removal, the dataset index was reset to maintain structure and consistency.

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**Data Type Correction**

Each feature was converted to its appropriate data type:

- Numeric → int / float
- Categorical → category
- Date/Time → datetime

This improves computational efficiency and analytical accuracy.

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**Text Standardization**

Categorical and text data were cleaned by:

- Removing extra spaces
- Converting to lowercase
- Fixing inconsistent spellings
- Removing special characters
- Standardizing category labels

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**Outlier Detection**

Outliers were identified using:

- Interquartile Range (IQR) method
- Z-score method
- Boxplot visualization

Extreme values were reviewed to ensure they were not valid business cases before treatment.

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**Data Consistency**

Business rules were applied to ensure logical correctness:

- No negative prices or quantities
- Ratings within valid range (1–5)
- Delivery time is realistic
- Distance values are non-negative

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**Data Validation**

After cleaning, the dataset was validated to ensure:

- No missing values remain
- No duplicate records exist
- Data types are correct
- All values are within valid ranges
- Dataset is ready for analysis

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**Workflow**

Raw Data → Inspection → Missing Values → Duplicates → Data Type Fix → Text Cleaning → Outlier Handling → Validation → Final Dataset

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**Benefits**

- Higher data quality
- Improved model performance
- Reduced noise in analysis
- More reliable insights
- Better visualization accuracy
- Strong foundation for machine learning

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# Data Quality Assessment

Data Quality Assessment is a critical step in any data science project as it ensures the dataset is reliable, consistent, and suitable for analysis and machine learning. For the Swiggy Food Delivery dataset, this stage focused on identifying missing values, duplicates, inconsistencies, invalid entries, and outliers to improve overall data reliability before modeling.

The main objectives were to evaluate completeness, detect missing or duplicate records, verify correct data types, ensure consistency in categorical variables, identify invalid or extreme values, and assess overall data integrity. These checks ensure that the dataset meets the requirements for accurate statistical analysis and predictive modeling.

The dataset was evaluated across key quality dimensions including completeness, accuracy, consistency, validity, uniqueness, and integrity. Each dimension plays an important role in ensuring that the data reflects real-world behavior and supports reliable business insights.

Duplicate records were checked to avoid redundancy in analysis. Exact duplicates and repeated transactional entries were identified and removed where necessary after verifying business relevance.

Data type validation ensured that each feature was stored in the correct format, such as numeric, categorical, or datetime. This improved computational efficiency and prevented errors during analysis.

Consistency checks were performed on categorical variables to standardize naming conventions, remove spelling variations, and ensure uniform formatting across features such as city names, restaurant categories, and cuisine types.

Range and validity checks were applied to ensure business logic compliance. For example, delivery time, order value, ratings, and quantity were verified to fall within acceptable operational limits.

Outlier detection was conducted using statistical methods such as IQR and Z-score analysis. Extreme values were carefully reviewed to distinguish between valid business cases and data errors.

Descriptive statistical profiling was used to understand data distribution, including mean, median, standard deviation, and skewness. This helped in identifying patterns and guiding transformation techniques.

Overall, the data quality assessment confirmed that while the dataset contained minor inconsistencies and missing values, it was suitable for further analysis after preprocessing. This step significantly improved the reliability of subsequent exploratory analysis, clustering, and machine learning models, ensuring accurate and meaningful business insights.

# Feature Engineering

Feature Engineering is a key step in transforming raw Swiggy transactional data into meaningful variables that improve model performance and business understanding. It focuses on creating, modifying, and selecting features that capture customer behavior, restaurant performance, and delivery efficiency.

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**Objectives**

- Improve model accuracy and performance
- Convert raw data into business-relevant features
- Capture hidden patterns in customer and restaurant behavior
- Reduce data complexity and redundancy
- Support segmentation and predictive analytics

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**Feature Categories**

**Numerical Features**

Price, Rating, Delivery Time, Distance, Order Value, Reviews
→ Checked for skewness, outliers, and distribution

**Categorical Features**

Restaurant, Cuisine, City, Payment Method, Order Status
→ Converted into numerical format using encoding techniques

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**Data Transformation**

- Standardization applied to normalize scale differences
- Log transformation used for skewed variables (e.g., order value, distance)
- Data types corrected (numeric, categorical, datetime)

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**Derived Features**

**Average Cost per Item**

Order Value ÷ Number of Items
→ Identifies customer spending behavior

**Delivery Efficiency**

Distance ÷ Delivery Time
→ Measures delivery performance

**Restaurant Popularity Score**

Rating × Number of Reviews
→ Identifies top-performing restaurants

**Customer Satisfaction Index**

Based on rating, delivery time, and order success
→ Measures overall service quality

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**Time-Based Features**

- Order Hour
- Day of Week
- Month
- Weekend Indicator
- Peak Hour Indicator

→ Helps analyze ordering patterns over time

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**Feature Scaling**

- Standardization (Z-score)
- Min-Max Scaling
- Robust Scaling (for outliers)

→ Ensures fair contribution of all variables in ML models

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**Feature Selection**

- Reduces overfitting
- Improves model speed
- Enhances interpretability

# Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a fundamental stage in the data science lifecycle that focuses on understanding the structure, patterns, and relationships within the dataset before applying machine learning models. In this Swiggy Food Delivery dataset, EDA was performed to analyze restaurant performance, customer behavior, pricing trends, delivery efficiency, and cuisine preferences, with the goal of generating actionable business insights.

The primary objective of EDA is to transform raw data into meaningful information by identifying distributions, detecting anomalies, and uncovering hidden patterns that support decision-making. The analysis begins with understanding dataset structure, including variables such as restaurant name, city, cuisine type, average rating, number of ratings, cost for two, delivery time, and other operational attributes. Initial data exploration ensures data quality by checking missing values, duplicates, data types, and summary statistics.

Univariate analysis was performed to study individual variables. Numerical features such as ratings, price, cost for two, and delivery time were analyzed using histograms, box plots, and density plots to understand distribution patterns, skewness, and the presence of outliers. Categorical variables such as cuisine, city, and restaurant type were analyzed using frequency distributions and count plots to identify dominant categories and customer preferences.

Bivariate analysis explored relationships between variables, such as rating vs cost, delivery time vs rating, and cuisine vs average rating. These relationships were visualized using scatter plots, regression lines, and box plots, revealing key insights such as the impact of delivery speed on customer satisfaction and the variation of ratings across cuisines. Multivariate analysis further examined interactions among multiple variables using correlation heatmaps and pair plots, helping identify strong and weak dependencies between features.

Correlation analysis quantified relationships between numerical variables, highlighting factors that influence customer ratings and pricing. Restaurant performance analysis combined multiple metrics such as ratings, reviews, and pricing to identify high-performing restaurants. Cuisine analysis revealed popular and high-rated food categories, while pricing analysis highlighted differences between budget and premium segments. Delivery analysis focused on delivery time distribution and its impact on customer satisfaction, while geographic analysis examined restaurant density and performance across cities.

Key business insights from EDA indicate that customer satisfaction is influenced by multiple factors including delivery efficiency, pricing strategy, and cuisine type. Premium pricing does not always guarantee higher ratings, and customer engagement varies significantly across cities and food categories. These insights are critical for improving operational efficiency, optimizing pricing strategies, and enhancing customer experience.

# Statistical Analysis

Statistical analysis was conducted to validate data quality, identify significant relationships among variables, and support data-driven business decisions. Both descriptive and inferential statistical techniques were applied to understand the distribution, variability, and correlation of key features within the Swiggy food delivery dataset.

**Descriptive Statistics**

Summary statistics, including mean, median, standard deviation, minimum, maximum, and quartiles, were calculated for numerical variables such as delivery time, order value, customer ratings, and delivery distance. These measures provided a comprehensive overview of the dataset and helped identify potential anomalies and data dispersion.

**Correlation Analysis**

Pearson's correlation coefficient was used to measure the linear relationships between numerical variables. The correlation matrix highlighted features that strongly influence delivery performance and customer satisfaction while identifying multicollinearity among predictors used in machine learning models.

**Outlier Detection**

Outliers were detected using the Interquartile Range (IQR) method and visualized through boxplots. Extreme observations were carefully evaluated and treated where necessary to improve model robustness without removing meaningful business information.

**Distribution Analysis**

The distribution of numerical variables was examined using histograms, density plots, and Q-Q plots. Normality tests, including the Shapiro–Wilk test, were performed to determine whether variables followed a normal distribution and to guide the selection of appropriate statistical techniques.

**Hypothesis Testing**

Inferential statistical methods were applied to evaluate business assumptions. Independent Sample T-Tests and One-Way ANOVA were used to determine whether significant differences existed between customer groups, restaurant categories, or delivery performance metrics. A significance level of $\boldsymbol{\alpha = 0.05}$ was adopted for all statistical tests.

**Statistical Findings**

The analysis identified meaningful relationships between delivery distance, order value, customer ratings, and delivery time. Statistical testing confirmed that several operational factors have a significant impact on delivery efficiency and customer experience, providing a strong analytical foundation for feature engineering, customer segmentation, predictive modeling, and business recommendations presented in subsequent sections.

# Hypothesis Testing

Hypothesis testing is a statistical technique used to determine whether observed patterns in the Swiggy dataset are statistically significant or occurred by chance. It supports data-driven decision-making by validating assumptions about customer behavior, restaurant performance, and delivery operations.

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**Objective**

The primary objective of hypothesis testing is to evaluate relationships and differences between variables that influence food delivery performance. A significance level of **$\alpha = 0.05$ (95% Confidence Level)** is used for all statistical tests.

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**Statistical Tests Performed**

**1. Independent Sample T-Test**

**Purpose:** Compare the average delivery time between two customer groups (e.g., Premium vs. Regular customers).

- **$H_0$:** There is no significant difference in the mean delivery time.
- **$H_1$:** There is a significant difference in the mean delivery time.

**Decision Rule:** Reject $H_0$ if **p-value $< 0.05$**.

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**2. One-Way ANOVA**

**Purpose:** Determine whether the average delivery time differs across multiple cities or restaurant categories.

- **$H_0$:** All group means are equal.
- **$H_1$:** At least one group mean differs significantly.

**Decision Rule:** Reject $H_0$ if **p-value $< 0.05$**.

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**3. Chi-Square Test of Independence**

**Purpose:** Evaluate whether customer ratings are associated with restaurant categories.

- **$H_0$:** The variables are independent.

- **$H_1$:** A significant association exists between the variables.

**Decision Rule:** Reject $H_0$ if **p-value < 0.05**.

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#### 4. Pearson Correlation Test

**Purpose:** Measure the strength and direction of the linear relationship between continuous variables such as delivery distance and delivery time.

- **$H_0$:** No linear correlation exists ($r = 0$).
- **$H_1$:** A significant linear correlation exists ($r \neq 0$).

Correlation values range from **-1 to +1**, indicating negative, positive, or no linear relationship.

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#### Interpretation

The results of these statistical tests provide evidence for identifying significant factors that influence delivery performance, customer satisfaction, and restaurant operations. Variables with statistically significant relationships are considered valuable for feature engineering, predictive modeling, and business decision-making, while non-significant variables may have limited analytical impact.

**Significance Level (α): 0.05**
**Confidence Level: 95%**

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# K-Means Clustering

K-Means Clustering is an unsupervised machine learning algorithm used to group similar observations based on their characteristics. In this project, K-Means was applied to identify distinct customer and restaurant behavior patterns within the Swiggy food delivery dataset. The objective was to discover meaningful segments that can support targeted marketing strategies, improve customer engagement, and optimize business operations.

**Methodology**

Before clustering, the dataset was preprocessed by handling missing values, removing outliers, and selecting relevant numerical features. Since K-Means is distance-based, feature scaling was performed using **StandardScaler** to ensure all variables contributed equally to cluster formation.

The optimal number of clusters was determined using two evaluation techniques:

- **Elbow Method:** Examined the Within-Cluster Sum of Squares (WCSS) across different values of *K* to identify the point where additional clusters provided diminishing improvements.
- **Silhouette Score:** Measured the quality of clustering by evaluating how well each observation matched its assigned cluster compared to other clusters.

Based on these evaluation metrics, the most appropriate number of clusters was selected and the K-Means algorithm was trained on the standardized dataset.

**Cluster Interpretation**

Each cluster represents a unique customer or restaurant segment with similar ordering behavior and operational characteristics. The segmentation helps identify high-value customers, budget-conscious users, premium restaurant groups, and other behavioral patterns that are useful for business decision-making.

Typical cluster characteristics include:

- High-value customers with larger average order values.
- Frequent customers placing regular food orders.
- Budget-oriented customers preferring lower-cost meals.
- Restaurants with consistently high customer ratings.
- Restaurants experiencing longer delivery durations.

## Regression Modeling

Regression modeling was used to predict the target variable using selected numerical and categorical features from the cleaned Swiggy dataset. The main objective was to understand key factors influencing the target and build a predictive model. Data preprocessing included handling missing values, encoding categorical variables, and scaling numerical features. Irrelevant and highly correlated features were removed to improve model stability. The dataset was split into 80% training and 20% testing sets for evaluation.

A Linear Regression model was chosen as the baseline due to its simplicity and interpretability. The model learns a linear relationship between input features and the target variable, represented as $\mathbf{Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_n X_n + \varepsilon}$, where Y is the predicted output and β values represent feature coefficients. The model was trained on the training set and tested on unseen data.

Performance was evaluated using MAE, MSE, RMSE, and $R^2$ Score. MAE measures average prediction error, RMSE penalizes larger errors, and $R^2$ Score indicates how well the model explains variance in the data. Residual analysis was performed to check prediction errors and validate model assumptions.

The model showed a reasonable fit and captured key relationships in the data. Feature coefficients helped identify the most influential variables. However, linear regression may not capture complex non-linear patterns. Future improvements include using Random Forest, Gradient Boosting, or XGBoost models. Overall, regression modeling provided a strong baseline for predictive analysis and business insights.

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## Model Evaluation

Model evaluation was performed to assess the accuracy and generalization of the regression model on the Swiggy dataset.
An 80:20 train-test split was used to ensure unbiased performance measurement.
The model was evaluated using MAE, MSE, RMSE, and $R^2$ Score.
MAE measured average prediction error in absolute terms.
MSE penalized larger errors more heavily to detect major deviations.
RMSE provided error in original units for better business interpretation.
$R^2$ Score indicated how well the model explained variance in the data.
Residual analysis confirmed that errors were randomly distributed.
Feature importance helped identify key influencing variables.
Overall, the model showed good predictive performance and generalization ability.

# Business Insights

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The analysis of the Swiggy Food Delivery dataset provides valuable insights into customer ordering behavior, restaurant performance, and delivery operations. These findings can help improve customer satisfaction, optimize business operations, and support data-driven decision-making.

**Customer Behavior**

- Customers show clear preferences for specific cuisines and highly rated restaurants, indicating that ratings significantly influence purchasing decisions.
- Higher order values are generally associated with premium restaurants and larger group orders.
- Peak ordering periods occur during lunch and dinner hours, resulting in increased demand and delivery workload.

**Restaurant Performance**

- Restaurants with consistently higher ratings receive more customer orders and generate greater overall revenue.
- A small percentage of restaurants contribute a significant share of total orders, highlighting the importance of maintaining strong partnerships with top-performing vendors.
- Cuisine popularity varies across customer segments, suggesting opportunities for targeted marketing campaigns.

**Delivery Operations**

- Delivery time increases as the delivery distance grows, making distance one of the most influential operational factors.
- Orders placed during peak hours experience relatively longer delivery times due to increased traffic and rider demand.
- Optimizing delivery routes and allocating riders based on demand forecasts can improve operational efficiency.

**Customer Segmentation**

- K-Means clustering identified distinct customer groups based on spending behavior and ordering patterns.
- High-value customers contribute a larger share of revenue and represent an important segment for loyalty and retention programs.

- Budget-conscious customers place frequent but lower-value orders, making them suitable for discount-based promotional strategies.

**Machine Learning Findings**

- The regression model successfully identified the key factors influencing the target variable, demonstrating the usefulness of predictive analytics in food delivery operations.
- Feature importance analysis indicates that variables related to order characteristics, customer behavior, and delivery conditions have the strongest impact on prediction performance.

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## Business Impact

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The analysis demonstrates that data-driven decision-making can significantly improve operational efficiency, customer experience, and business profitability. By leveraging customer segmentation, predictive modeling, and operational analytics, Swiggy can optimize resource allocation, enhance delivery performance, personalize marketing campaigns, and strengthen long-term customer engagement.

**Business Recommendations**

- Optimize delivery using AI-based routing and demand forecasting to reduce delays.
- Improve customer retention through personalized loyalty programs and targeted offers.
- Strengthen restaurant partnerships by promoting high-performing vendors and improving low performers.
- Use dynamic pricing and segmented promotions to increase marketing efficiency and ROI.
- Apply customer segmentation (K-Means) for personalized marketing and recommendations.
- Deploy predictive models to estimate delivery time/order value for better planning.
- Build real-time dashboards to track KPIs like orders, delivery time, and revenue.
- Continuously retrain models to maintain accuracy as data evolves.

**Expected Impact**

Improved delivery efficiency, higher customer satisfaction, better retention, optimized costs, and increased overall business profitability.

## Limitations

This project is based on a single Swiggy dataset, which may not fully represent all customer behaviors, restaurant performance, or delivery patterns across different regions and time periods. The accuracy of insights depends on the completeness and quality of the available data.

The machine learning models are limited to the features present in the dataset. Key real-world factors such as weather, traffic, promotions, delivery partner availability, and customer demographics were not included, which may affect prediction accuracy.

K-Means clustering assumes spherical clusters and requires a predefined number of clusters. Although methods like Elbow and Silhouette Score were used, different clustering techniques may produce different segmentations.

The models are trained on historical data, so changes in customer behavior or business conditions over time may reduce performance unless models are regularly updated.

Finally, the insights and recommendations are data-driven but should be validated with domain expertise and business constraints before real-world implementation.

## Future Scope

This project provides a strong base for food delivery analytics, but it can be further improved by adding real-time data processing to monitor operations and support faster decision-making. Integrating cloud-based pipelines and dashboards would allow live tracking of key business metrics.

The machine learning models can be enhanced using advanced algorithms like XGBoost, LightGBM, and neural networks to improve accuracy. Automated hyperparameter tuning and model monitoring can also be added for better performance and reliability.

Customer segmentation can be improved using advanced clustering techniques such as DBSCAN or Gaussian Mixture Models, along with customer behavior and demographic data to enable better personalization and marketing strategies.

Future work can also include recommendation systems for food and restaurants, along with predictive models for customer churn, demand forecasting, and delivery time estimation to improve efficiency and user experience.

Finally, deploying the project as a web application using tools like Streamlit or Flask, along with MLOps practices, can make it a scalable, production-ready system for real-world business use.

## Conclusion

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**In this project, we analyzed Swiggy food delivery data from 28 Indian cities covering the period from January to August 2025. The dataset contained nearly 200,000 dish listings across hundreds of restaurants.**

We performed detailed Exploratory Data Analysis to understand pricing patterns, ratings, popular categories, and city-wise differences. Additionally, we applied **K-Means Clustering** to segment dishes and built a **Linear Regression model** to predict dish prices.

*Key Takeaways:*

- Price and Rating have very weak correlation.
- Many dishes lack sufficient customer ratings.
- Dishes can be meaningfully grouped using clustering techniques.
- Predicting price using only Rating and Rating Count has limited accuracy.

*Final Thoughts:*

**This analysis provides useful insights into pricing patterns, customer preferences, and restaurant performance on Swiggy. The findings can help restaurants optimize their menus and pricing strategies, while also helping customers discover better value options.**

**Overall, this project demonstrates the complete Data Science workflow — from data cleaning and exploration to machine learning and business recommendations.**

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## Appendix

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**A. Project Repository Structure**

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Swiggy-Food-Delivery-Data-Analysis-ML/
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├── data/
|   ├── raw/
|   └── processed/
├── notebooks/
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├── src/

├── models/

├── reports/

├── images/

├── requirements.txt

├── README.md

└── LICENSE

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**B. Software & Libraries**

- Python 3.12
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- Scikit-learn
- SciPy
- Statsmodels
- XGBoost (Optional)
- SHAP
- Joblib

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**C. Machine Learning Workflow**

Raw Dataset → Data Cleaning → Feature Engineering → Exploratory Data Analysis → Statistical Analysis → Feature Scaling → Train-Test Split → Model Training → Hyperparameter Tuning → Model Evaluation → Business Insights → Recommendations

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**D. Key Evaluation Metrics**

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- $R^2$ Score
- Adjusted $R^2$
- Silhouette Score (K-Means)

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**E. Project Deliverables**

- Cleaned Dataset
- Exploratory Data Analysis Notebook
- Statistical Analysis Report
- Feature Engineering Pipeline
- K-Means Clustering Model
- Regression Model
- Model Evaluation Report
- Business Insights & Recommendations
- Professional GitHub Repository
- Final Project Documentation

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**F. References**


1. Swiggy Public Dataset (Project Dataset)
2. Scikit-learn Documentation — https://scikit-learn.org/
3. Pandas Documentation — https://pandas.pydata.org/
4. NumPy Documentation — https://numpy.org/
5. SciPy Documentation — https://scipy.org/
6. Statsmodels Documentation — https://www.statsmodels.org/
7. Matplotlib Documentation — https://matplotlib.org/

8. Plotly Documentation — https://plotly.com/python/

9. CRISP-DM: Cross-Industry Standard Process for Data Mining

10. Python Software Foundation — https://www.python.org/

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**G. Author Information**

**Project:** Swiggy Food Delivery Data Analysis & Machine Learning (2025)

**Domain:** Data Science | Machine Learning | Business Analytics

**Project Type:** End-to-End Industry Case Study

**Workflow:** Data Cleaning → Feature Engineering → Exploratory Data Analysis → Statistical Analysis → Machine Learning → Business Insights → Strategic Recommendations

**Repository:** GitHub Portfolio Project

**Project: Data Science Swiggy Dataset Analysis.ipynb**